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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	1	1	3	3	3	2	4	3	4	3	2	2	2	3	4	1	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	3	4	2	3	4	2	4	2	4	2	2	3	2	3	3	1	1	2
Q.	41	42	43	44	45															
A.	4	4	3	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	2	2	4	4	3	2	1	1	1	1	3	4	4	2	1	4	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	4	1	3	4	1	2	3	2	2	4	2	2	4	2	2	1	3	2
Q.	86	87	88	89	90															
A.	1	2	4	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	2	2	1	2	3	4	3	1	1	1	3	1	2	2	1	3	3	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	1	2	4	3	2	4	3	2	2	3	4	3	3	2	2	1	4	4	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	4	1	4	2	2	4	2	2	4	4	4	4	1	1	4	4	4	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	1	1	3	3	2	2	3	4	2	1	2	3	1	2	4	3	1	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	1	4	4	1	3	2	4	3	1										

SOLUTIONS

PHYSICS

1) $Q = \frac{\omega}{\Delta\omega} = \frac{\omega L}{R} \Rightarrow \Delta\omega = \frac{R}{L} = \frac{80}{8} = 10$ Hz

$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 50 \times 10^{-6}}} = 50$

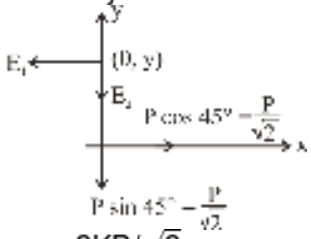
$\omega_L = \omega - \frac{\Delta\omega}{2} = 50 - 5 = 45$ rad/s

$\omega_H = \omega + \frac{\Delta\omega}{2} = 50 + 5 = 55$ rad/s

2) For X : Energy = 200 × 7.4 = 1480 MeV
For A : Energy = 120 × 8.2 = 984 MeV
For B : Energy = 80 × 8.2 = 656 MeV
 Energy released = (984 + 656) – 1480
= 1640 – 1480 = 160 MeV

3) $E_P = \sqrt{E_1^2 + E_2^2}$

$E_1 = \frac{KP/\sqrt{2}}{y^3}$



$E_2 = \frac{2KP/\sqrt{2}}{y^3}$

$E_P = \frac{KP\sqrt{5}}{\sqrt{2}y^3}$

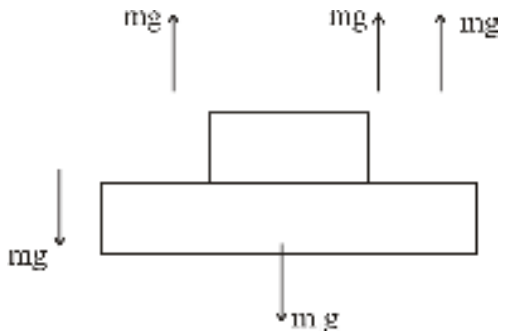
4) Energy supplied by battery $U = C_{net} V^2$
(double of stored energy)

5)

$\delta_{min} = 2i - A \Rightarrow A = 2i - \delta_{min}$
$= 2 \times 45 - 30 = 60°$.

$\mu = \frac{\sin\frac{(A+\delta min)}{2}}{\sin A/2} = \frac{\sin\left(\frac{60+30}{2}\right)}{\sin\left(\frac{60}{2}\right)} = \frac{1 \times 2}{\sqrt{2}} = \sqrt{2}$

$\mu = \sqrt{2} \Rightarrow \sin\theta_C = \frac{1}{\mu gm} = \frac{1}{\sqrt{2}}$
$\Rightarrow \theta_C = 45°$



6)
$3mg = mg + m_1g$
$m_1 = 2m$

7) $V^0 = \sqrt{\frac{GM}{r}} \quad \Rightarrow \quad V^0 \propto \frac{1}{\sqrt{r}}$
$\frac{V_1}{V_2} = \sqrt{\frac{r_2}{r_1}}$
$\frac{V}{V_2} = \sqrt{\frac{4R}{R}}$, $V_2 = \frac{V}{2}$

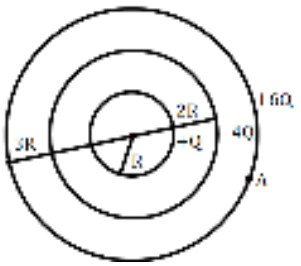
8)

With constant positive acceleration, speed will increase when velocity is positive, speed will decrease, if velocity is negative.

Similarly, with constant negative acceleration, speed will increase, if velocity is negative and speed will decrease, if velocity is positive.
Hence, A → p, q, B → p, q, C → r, D → p, q.

9)

According first law
Q = ΔU + w
w = +ve in all cases
ΔU = Q – w
Also, ΔU = 0(for Isothermal)

10) 

$$\left|\vec{E}_A\right| = \frac{K(Q)}{(3R)^2} + \frac{K(-4Q)}{(3R)^2} + \frac{K(6Q)}{(3R)^2}$$

$$\left|\vec{E}_A\right| = \frac{KQ}{9R^2}[1-4+6]$$

$$|E| = \frac{3Q}{4\pi\varepsilon_0 \times 9R^2}$$

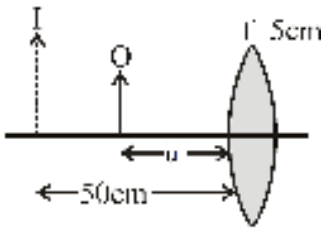
$$|E| = \frac{Q}{12\pi\varepsilon_0 R^2}$$

11)

$m_1x_1 + m_2x_2 = 0$
$5(x + y) + 20x = 0$
$5x + 20 + 20x = 0$
$25x = -20$
$x = -\frac{4}{5}$
Dist. of dog from shore
$10 - \left(4 - \frac{4}{5}\right)$
$10 - \frac{16}{5} \Rightarrow \frac{50-16}{5} = \frac{34}{5} = 6.8$ m

12)

LC = $\frac{1\text{mm}}{100}$ = 0.01 mm,
[Let the thickness of each sheet = t]
Case-I : For single sheet
$t = R_1 = 2 + 12(0.01)$ mm – ZE
$\Rightarrow t = 2.12$ – ZE ..... (1)
Case-II For two sheets together
$2t = R_2 = 4 + 10(0.01)$ mm – ZE
$2t = 4.10$ – ZE ..... (2)
From (1) and (2)
$2(2.12) - 2ZE = 4.10 - ZE$
$\Rightarrow$ ZE = 4.24 – 4.10 = + 0.14 mm



13)
u = ?
v = –50 cm
f = 5 cm
$-\frac{1}{50} - \frac{1}{u} = \frac{1}{5}$

$u = -\frac{50}{11}$

M.P. $= \frac{D}{|u|} = \frac{25}{50} \times 11 = 5.5$

14) For three resonance :

$\frac{5\lambda}{4} \leqslant 1.00\text{mm}$

$\Rightarrow \lambda \leq 0.80$ m

$\Rightarrow \frac{f}{v} \geqslant \frac{1}{0.80}$ or $f \geq \frac{340}{0.80}$

$\Rightarrow f \geq 425$ Hz

15) This question is based on work done in polytropic process $\left[W = \frac{nR}{1-x}[T_2 - T_1]\right]$

$PV^{-2} = a$

'x = –2'

$W = \frac{nR}{1-(-2)}(T_2 - T_1)$

$W = \frac{R}{3}(T_2 - T_1)$

16) From principle of calorimetry, Heat lost = Heat gained

Heat lost by metal block $\Rightarrow Q_1 = 100 \times 0.10\ (120 - 0°) = 1200$ cal

Heat required to melt 'x' g of ice = $Q_2 = 80x$

Heat required to warm (100 + 20 + x) g water from 0°C to 0°C = 0

So at equilibrium $Q_1 = 80x$

$\Rightarrow x = \frac{1200}{80} = 15\text{g}$.

17) By work energy theorem

$W_F + W_{\text{friction}} = \frac{1}{2}m\left[v^2 - u^2\right]$

$(40 \times 5) + W_{\text{friction}} = \frac{1}{2}(2)\left[(12)^2 - 0\right]$

$W_{\text{friction}} = 144 - 200 = -56$ J

18)

By Einstein's equation

$\frac{hc}{\lambda} = 3eV_0 + \frac{hc}{\lambda_0}$ and $\frac{hc}{2\lambda} = eV_0 + \frac{hc}{\lambda_0}$

$\Rightarrow \frac{hc}{\lambda} - \frac{hc}{\lambda_0} = 3\left(\frac{hc}{2\lambda} - \frac{hc}{\lambda_0}\right)$

$\Rightarrow \frac{2hc}{\lambda_0} = \frac{hc}{2\lambda} \Rightarrow \lambda_0 = 4\lambda$

19)

$F_{magnetic} > F_{Electric}$ then charge will move in –ve z direction
if $q V_0 B > q E$
$V_0 > E/B$

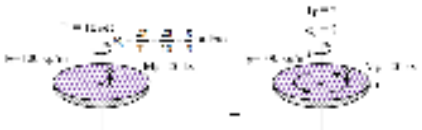
$V^0 > \frac{4 \times 10^3}{0.1}$ $V^0 > 4 \times 10^4$ m/s Hence $V^0 = 5 \times 10^4$ m/s

20)

To keep the disc horizontal
Net torque = 0 (about point A)
Mg(R) – T(2R) = 0
MgR – mg (2R) = 0 (T = mg)

$\boxed{m = \frac{M}{2}}$

21)



L_{system} = constant
Iω = constant
$I_i\omega_i = I_f\omega f$
$(I_P + I_B)\omega_i = (I_P + I_P)\omega_f$

$$\left[100 + 20 \times (0)^2\right]\left(\frac{\pi}{5}\right) = \left[100 + 20 \times (2)^2\right](\omega_f)$$

$$[100 + 0]\left(\frac{\pi}{5}\right) = [100 + 80](\omega_f)$$

$$[100]\left(\frac{\pi}{5}\right) = [180](\omega_f)$$

$20\pi = (180)\omega_f$

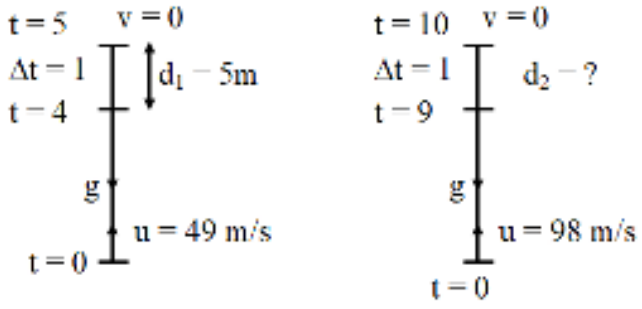
$$\frac{20\pi}{180} = \omega_f$$

$\omega_f = \frac{\pi}{9}$ rad/sec

$T_f = \frac{2\pi}{\omega} = \frac{2\pi}{(\pi/9)} = 18$ sec

22) $7 \times 30 = 3 \times 10 + W \times 60$
w = 3N

23)



We have to calculate d_2 = ?
Distance travelled in last second during upwards motion is always equals to distance travelled in 1st second of down ward journey. it does not depend on initial velocity of upward motion.
It is calculated by

$$d = \frac{1}{2} \times g \times (\Delta t)^2$$

$$d = \frac{1}{2} \times g \times (1)^2$$

d = 5 m.

24) $G = \frac{RS}{R-S}$

$$G_1 = \frac{(3300)(80)}{(3300-80)} \approx 82\Omega$$

$$G_2 = \frac{(5000)(80)}{(5000-80)} \approx 81\Omega$$

$$G = \frac{G_1 + G_2}{2} \approx 80\Omega$$

25)

Stress = Y × strain

$$\frac{mg}{A} = Y \times \text{strain}$$

$m \times 10 = 2 \times 10^{11} \times 10^{-3} \times 3 \times 10^{-6}$

m = 60 kg

26) Let $v = kg^x \lambda^y \rho^z$
Now by substituting the dimensions of each quantities and equating the powers of M, L and T we get z = 0, x = 1/2, y = 1/2

27) $I_A \cos^2 30º = I_B \cos^2 60º$

$$\Rightarrow \frac{I_A}{I_B} = \frac{1}{3}$$

28)

$P_{Bottom} = P_0 + \rho gH$

4 atm = 1 atm + ρgH

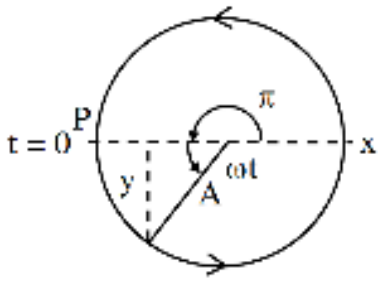
$\rho gH = 3 \text{ atm} = 3 \times 10^5$ Pa

H = 30 m

$v = \sqrt{2gH} = \sqrt{600}$ m/s

29) a. Uniform cross-section of bridge wires ensures that resistance per unit length is constant
b. Resistance which are nearly equal ensures null point near the center of bridge wire leading to increase in sensitivity.
c. Galvanometer of low resistance aids high sensitivity.

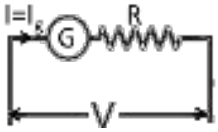
d. Using thicker wire decrease the potential gradient which in turn make null point less sharp.



30)
y = Asin(ωt + π)
y = –Asinωt

31)

Given here
G = 0.1 kΩ = 100Ω, I_g = 100 μA= 100 × 10^{-6}, V = 0.1V



From circuit, V =(G + R)I_g

$$R = \frac{V}{I_g} - G$$

$$R = \frac{0.1}{100 \times 10^{-6}} - 100$$

R = 1000 – 100 = 900 Ω

32) $\oint B.d\ell = \mu_0 I_{enc}$

Case 1: Inside the Wire (r < R)

$$I_{enc} = i\frac{r^2}{R^2}$$

$\oint B.d\ell = \mu_0 i\frac{r^2}{R^2}$ (Quadratic increase)

Case 2: Outside the Wire (r ≥ R)

I_{enc} = i

$\oint B.d\ell = \mu_0 i$ (Constant)

Graphical Analysis

A. r < R : Quadratic increase.
B. r ≥ R : Constant.

33) $$\left(\frac{\lambda_{Lyman}}{\lambda_{Balmer}}\right)_{max} = \frac{\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}{\left(\frac{1}{1^2} - \frac{1}{2^2}\right)} = \frac{5/36}{3/4} = \frac{5}{27}$$

34) Rate of heat loss

$= e\sigma_A(T^4 - T_0^4)$
$= 0.4 \times 5.67 \times 10^{-8} \times 200 \times 10^{-4}$
$\times [(800)^4 - (300)^4]$
$= 5.67 \times 80 \times 10^{-4} \times (4096 - 81)$
$= 182.12$ J/s ≈ 182 J/s

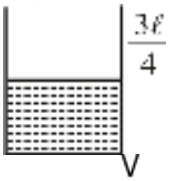
35)

Without zener diode : $V_{2k\Omega} = \frac{10}{3.5} \times 2 = 5.7V$
Thus zener will not conduct as $V_{2k\Omega} < V_Z$
So current through $2k\Omega = \frac{10}{3.5k\Omega} = 2.85$ mA.

36)

Using force balance method :
For capillary A : $(2\pi a)T\cos\theta = \pi a^2 h\rho g$
$\Rightarrow h_A = \frac{2T\cos\theta}{a\rho g}$ ...(1)
For capillary B : $(4a)T\cos\theta = a^2 h\rho g$
$\Rightarrow h_B = \frac{4T\cos\theta}{a\rho g}$ ....(2)
From (1) and (2) $\frac{h_A}{h_B} = \frac{1}{2}$

37) $\frac{V}{4\ell} = 220$ Hz

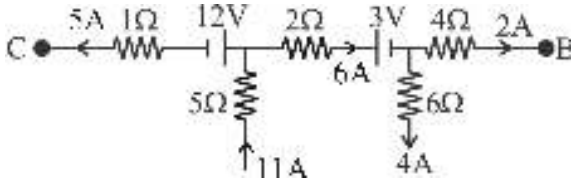


n' = 3 $\frac{V}{4 \times \frac{3\ell}{4}} = \frac{V}{\ell} = 880$ Hz

38)

In reverse bias current is mainly due to minority charge carriers which can easily be affected by light.

39)



$V_C - V_B = 1(-5) - 12 + 2(+6) + 3 + 4(+2)$

= 6 Volt

40)

$$P_{maximum} = \frac{E^2}{4r} = \frac{(4)^2}{4 \times 1} = 4W$$

so power should be less than or equal to 4W.

41)

$$r = \frac{\text{depth}}{\sqrt{\mu^2 - 1}} = \frac{2\sqrt{7}}{\sqrt{(4/3)^2 - 1}}$$

r = 6 m

diameter = 12 m

42)

$$\left|\vec{A} - \vec{B}\right| = \sqrt{A^2 + B^2 - 2AB\cos\theta}$$

Given:

$$A = 2, B = 4, \theta = 60^\circ$$

$$\left|\vec{A} - \vec{B}\right| = \sqrt{2^2 + 4^2 - 2\,(2)\,(4)\cos 60^\circ}$$

$$\left|\vec{A} - \vec{B}\right| = \sqrt{4 + 16 - 16\left(\frac{1}{2}\right)}$$

$$= \sqrt{20 - 8} = \sqrt{12} = 2\sqrt{3}$$

43) de-Broglie wavelength

$$\lambda = \frac{h}{mv_{rms}} = \frac{h}{\sqrt{3mkT}}$$

$$\frac{\lambda_H}{\lambda_{He}} = \sqrt{\frac{m_{He}T_{He}}{m_H T_H}} = \sqrt{\frac{4(273+127)}{2(273+27)}} = \sqrt{\frac{8}{3}}$$

44)

$$E = E_0 \sin(\omega t - kx) \text{ and } B = B_0 \sin(\omega t - kx)$$

$$U = \frac{1}{2}\varepsilon_0 E^2 + \frac{B^2}{2\mu_0}$$

$$U = \frac{1}{2}\varepsilon_0 E_0^2 \sin^2(\omega t - kx) + \frac{B_0^2}{2\mu_0}\sin^2(\omega t - kx)$$

Average value of $\sin^2(\omega t - kx) = \frac{1}{2}$

$$\therefore U_{average} = \frac{1}{4}\varepsilon_0 E_0^2 + \frac{B_2}{4\mu_0}$$

45) $L = \frac{\mu_0 N^2 A}{\ell} = \frac{\mu_0 N^2 \pi R^2}{\ell}$

$\ell_w = N(2\pi R) \Rightarrow NR = \frac{\ell_w}{2\pi}$

$L = \frac{\mu_0 \pi}{\ell} \times \left(\frac{\ell_w^2}{4\pi^2}\right)$

CHEMISTRY

46)

$\Delta G = \Delta H - T\Delta S$
$\Delta H = \Delta G + T\Delta S$
$= 20000 + 200 \times (-20)$
$= 16000$ J/mol
Using Kirchoff's equation,
$\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$
$\Delta H_2 = \Delta H_1 + \Delta C_p (T_2 - T_1)$
$= 16000 + 20 (400 - 200)$
$= 20000$ J/mol
$= 20$ kJ/mol.

47)

$\Delta T_f = k_f \times i \times m$

$\Delta T_f = 1.86 \times 1 \frac{\left(\frac{5}{60} + \frac{10}{180}\right)}{85} \times 1000$

$\Delta T_f = 1.86 \times \frac{25}{180} \times \frac{100}{85} = 3.03$

$T_f^0 - T_f = 3.03$
$T_f = -3.03$ °C

48)

H_2CO_3 and HCO_3^- works as a buffer solution for blood.

49)

Lower will be (n+□), lower will be Energy of e^-

50) $\mu_L = \left(\sqrt{\ell(\ell+1)}\right) \frac{h}{2\pi}$

$\mu_L = \left(\sqrt{3(3+1)}\right) \hbar$

$\mu_L = \sqrt{12}\hbar$

51) At t = 3 min, according to graph
$[H_2]$ increases first ie, H_2 was added to the equilibrium

52)

1 ppm means component present in 10^6 part of solution

53)

$C_t = C_0\, e^{-kt}$

54)

$E_a = 23 - 20 = 3$ Kcal
(Q → C)

55)

Slope = $-\frac{E_a}{R} = -\frac{20}{5}$
$E_a = 4R = 4 \times 2$ Cal

56) $(MnO_4^{-2} \rightarrow MnO_4^- + 1e^-) \times 2$
$4H^+ + MnO_4^{2-} \rightarrow MnO_2 + 2H_2O$
$+2e^-$
$3MnO_4^{2-} \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$
÷ by 3
$1MnO_4^{2-} \rightarrow \frac{2}{3}MnO_4^- + \frac{1}{3}MnO_2 + 2H_2O$

57)

NCERT Pg. 275

58) $\Delta H_r^0 = (\Delta H_f^0)_P - (\Delta H_f^0)_R$

59)

(i) $NaCl_{(s)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)}$
$\Delta rH° = \Delta H°_{solution}$
(ii) $NaCl_{(s)} \rightarrow Na^+_{(g)} + Cl^-_{(g)}$
$\Delta rH° = \Delta H°_{lattice}$
(iii) $Na_{(s)} \rightarrow Na_{(g)}$
$\Delta rH° = \Delta H°_{atomisation}$

(iv) $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\ell)}$

$\Delta_r H^\circ = \Delta H^\circ_{\text{formation}}$

60) $3CaBr_2 + 2K_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6KBr$

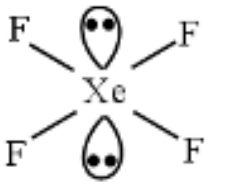
t = 0	0.5 mol	0.2 mol (L.R.)	0	0
After reaction	0.5 – 0.3 = 0.2 mol	0	0.1 mol	0.6 mol

61) SO_3^{-2} has dπ–pπ Bonding

⇒ Rest species do not have d-orbital.

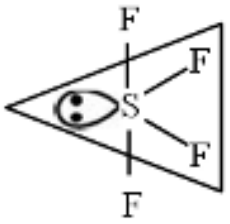
62) Polarising power : $M^{+1} < M^{+2} < M^{+3}$

63) (A)



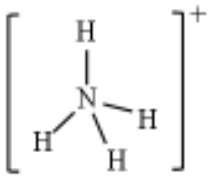
Square planer

(B) SF_4



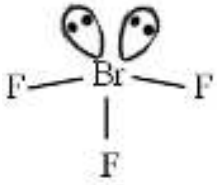
See-Saw

(C) NH_4^+



Tetrahedral

(D) BrF_3



Bent T- Shaped

64)

$[Mn(CO)_6]^+$ has minimum synergic bonding
Hence less tendency to acquire electron pair in ABMO of CO. more bond strength of C & O.

65)

Trans–$[CoCl_2(en)_2]^+$ is not optically active. In trans isomer plane of symmetry is present.

66)

(A) $[CrF_6]^{4-}$ has Cr^{2+} (d^4) with weak field F- : 4 unpaired electrons (iii).
(B) $[MnF_6]^{4-}$ has Mn^{2+} (d_5) with weak field F- : 5 unpaired electrons (iv).
(C) $[Cr(CN)_6]^{4-}$ has Cr^{2+} (d_4) with strong field CN- : 2 unpaired electrons (ii).
(D) $[Mn(CN)_6]^{4-}$ has Mn^{2+} (d_5) with strong field CN- : 1 unpaired electron (i).

67)

$NaCl + H_2SO_4 + K_2Cr_2O_7$

$\downarrow$

$CrO_2Cl_2 \xrightarrow{NaOH} Na_2CrO_4$

(Orange red Vopour)

68)

$[Cr(H_2O)_6]Cl_3$ will give 3 moles of AgCl on addition of excess of $AgNO_3$.

69)

Np show highest O.S. up to +7

70)

$AlCl_3 + 6H_2O \rightarrow [Al(H_2O)_6]^{+3}(aq) + 3Cl^-(aq)$

71)

A. $(CH_3)_3SiOH$ is more acidic than $(CH_3)_3COH$ due to higher Si-O bond polarity and stabilization of the conjugate base by silicon.

B. Silicon monoxide (SiO), unlike stable CO, is unstable in monomeric form under normal conditions and tends to polymerize.

C. The B-F bond in BF_3 is shorter than in BF_4^- due to pπ-pπ back bonding in BF_3, giving it

partial double bond character.

Therefore, the statement "None of these" is incorrect because the first three statements are actually correct.

Final Answer - Option (4)

72)

O → Min EA
S → Max EA

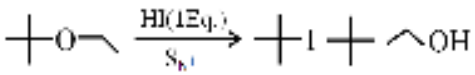
73) Cu^{+} ions spontaneously react to form both Cu(metallic copper) and Cu^{2+} ions. $2Cu^{+}(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$. So it is true statement.
Hydrogen sulfide (H_2S) does not impart a purple color in the sodium nitroprusside test, So it is true statement. The oxidation state of iron does not change in the reaction between sulphide ions and sodium nitroprusside, specifically from +2 to +7. This reaction results in the formation of a purple-colored complex. So it is true statement.

74)

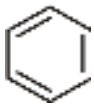
Pb is not a transition metal and Borax bead test is not performed by non transition metal.

75)

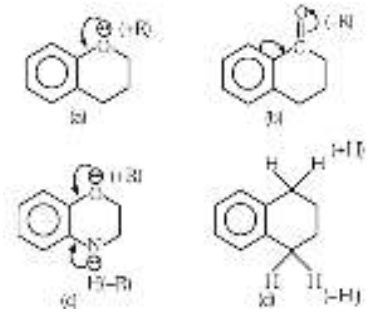
If HCl is not added before passing H_2S in the second group, it may result in the precipitation of Group IV cations
This causes confusion and incorrect group identification.

76) 

77)

A is 

78) Reactivity towards electrophilic aromatic substitution reaction (ESR) a density in ring.



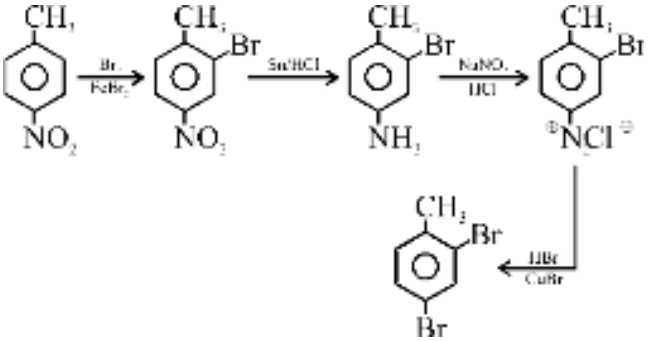
(2) c > a > d > b

79)

Explain the Question : Determining product (D) starting from p-nitrotoluene through bromination, reduction, and Sandmeyer reaction.

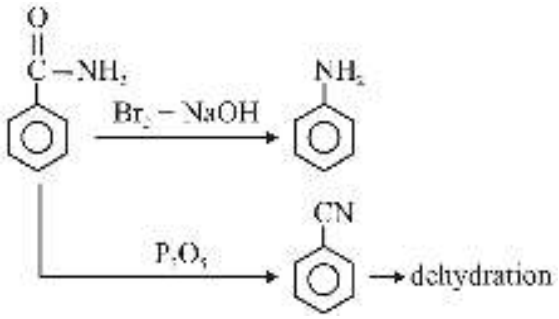
Concept : This question is based on Directing effects, Reduction of nitro groups, Diazotization and Sandmeyer reaction.

Solution :



Final answer : Option (2)

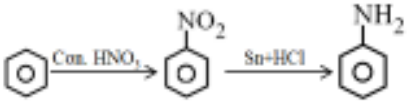
80)



81)

Name reaction based question

82)

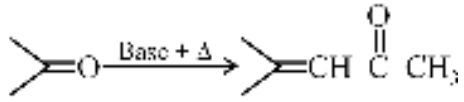


83)

Vitamin "C" → Ascorbic acid

84)

Aldol condensation



85)

Rate of NAR $\propto \frac{1}{\text{Steric crowding around } \angle C=O}$

86)

Ncert-Class 11th,Part-2, Pg.No: 273 Edition-2023-24

87)

Data base Question

88)

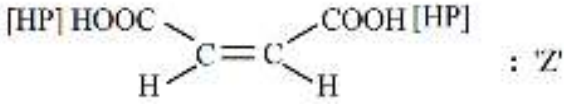
$K_b \propto \frac{+M}{+I}$

89)

Rate of S_{N1} ∝ stability of C^+

90)

Explaining the Question :-
Identity 'Z' isomer.
Concept :-
If High priority group [HP] according to CIP rule are on same side of geometrical isomer than assign 'Z'.
Solution :-



Final Answer :- Option **(4)**

91)

NCERT-XII, Pg. # 167

92)

NCERT-XII, Pg. # 171, 172, 174

93)

NCERT-XII, Pg. # 166, 167

94) NCERT 12th, Page No. # 184

95)

NCERT 12th, Page No. # 179,180,181

96)

NCERT-XII, Pg. # 153

97)

Module-4, Pg. # 121

98) NCERT-XII, Pg. # 38

99) NCERT-XI, Pg # 49

100)

NCERT, Pg. # 44

101) NCERT, Pg. # 46

102) NCERT, Pg. # 44

103) NCERT Pg # 82, 83

104)

NCERT, Pg. # 80

105) Module-4, Pg. # 88

106)

NCERT, Pg. # 200

107) NCERT, Pg. # 198

108)

NCERT, Pg. # 197

109) NCERT Pg.#194, 198.

110) NCERT Pg. # 185

111) NCERT, Pg. # 188

112)

NCERT, Pg. # 189

113) NCERT-XI, Pg. # 207

114) NCERT Pg. # 213

115) NCERT Pg # 212

116) NCERT (XII) ; Pg. # 319

117) NCERT (XII) ; Page No. # 317 (E)
NCERT (XII) ; Page No. # 316 (H)

118) NCERT (XII) ; Page No. # 231

119)

NCERT, Pg. # 241

120) NCERT Pg.#243

121) NCERT 337

122) NCERT XI Pg.#240,241

123) NCERT Pg.#227

124)

NCERT, Pg. # 221

125)

NCERT, Pg. # 224

126) NCERT Pg. # 31

127) NCERT PAGE (ENGLISH): 38

128) NCERT, Page # 27

129) NCERT, Pg. # 44

130) NCERT, Pg. # 43

131)

NCERT, Pg. # 42

132) NCERT Pg. # 141

133) NCERT Pg. # 142 New

134) NCERT Pg. # 130-133

135) NCERT Pg. # 121

136) **Asking About :-** value of Entropy change
Concept :- Entropy and change of entropy (ΔS)
Solution :-
1. $N_2 + 2H_2 \rightarrow 2NH_3$, moles decreases (so that disorder decreases $\Delta S < O$)
2. $H_{2(g,\ 10atm)} \rightarrow H_{(g,10atm)}$
(The gas expands from high pr. to low pressure, volume increasivy so that discrder Increases $\Delta S > O$)
3. $Fe_{(s)\ T= 800k)} \rightarrow Fe_{(s)(T = 400k)}$
T ↓ S ↓ DS < 0
4. $H_2O \rightarrow H_2O(l)$
Gas conderses to liquid, derisive disorder as parties be come more ordered
$\Delta S < 0$
Correct Ans. → Option 2

137)

NCERT-XI Pg#33

138)

Question Explanation :
The question is asking prokaryotic cell structure, specifically the organization of genetic material in *Escherichia coli* (*E. coli*).

Concept :
This question is based on Organization of genetic material in prokaryotic cells.

Solution :
Assertion: The genomic DNA of *E. coli* is considered "naked" because it is not enclosed within a membrane-bound nucleus.
Reason: Prokaryotes lack a nuclear envelope, which is a defining characteristic distinguishing them from eukaryotes.
Why Reason is not the correct explanation: While the lack of a nuclear envelope in prokaryotes is true, it does not directly explain why *E. coli* genomic DNA is considered "naked."

Final Answer :
Option (2)

139) NCERT-XI Pg.#23

140)

Solution:

The correct answer is:

4. It was a phylogenetic classification.

Bentham and Hooker's classification was based on gross morphology and natural classification but was not phylogenetic (i.e., it did not reflect evolutionary relationships).

141)

A. Agaricus (a genus of mushrooms) is a Basidiomycete and produces basidia for sexual reproduction.
B. Ascomycetes produce ascus for sexual reproduction, where ascospores are formed.

142)

Not in any kingdom

143) NCERT XI_Page No.86

144)

NCERT XI, Page 87

145)

NCERT Pg. No. 77

146)

Concept:

A. Telomeres are repetitive DNA sequences at the ends of chromosomes that protect them from damage.

B. Telomeres shorten with each cell division, and this shortening is linked to aging.

Solution:

A. Telomere shortening is a well-established biological marker of aging.

147)

Explain Question: The question asks to select the correct option.

Concept:Plastid

Solution: The correct statement is:

Chloroplasts are found in plant cells but not in prokaryotic or animal cells.

So, the correct answer is: 4.

Final answer: 4

148) NCERT XI, Eng. Pg.# 170 (Significance of Meiosis)
हिन्दी पेज # 170 (आखिरी पेराग्राफ)

149)

NCERT XI Pg. # 146

150) NCERT Pg. No. # 143

151)

NCERT-XI, Pg # 248

152) NCERT XI Pg.# 173

153)

NCERT, Pg. # 222

154)

NCERT Pg. # 209 (NCERT-2022)

155) NCERT Pg. No. # 152
aerobic respiration is found due to presence of mitochondria and includes Krebs cycle and link reaction

156)

NCERT Pg # 156

157) NCERT (XI) Pg. # 221

158) NCERT-XII, Pg. # 22, 23

159) NCERT-XII, Pg. # 11

160) NCERT-XII, Pg. No. # 18

161) NCERT-XII, Pg # 69

162) NCERT XII Pg. No. # 74, 75

163) NCERT XII Pg. No. # 63

164) NCERT XII Pg. No. # 160, 62

165) NCERT XII pg no.72

166) NCERT-XII, Pg # 106

167) NCERT-XII, Pg # 101

168)

NCERT-XII, Pg. # 110

169) NCERT-XII, Pg. # 109

170) NCERT-XII, Pg. # 114

171) NCERT-XII, Pg. # 112

172) NCERT-XII, Pg. # 97

173) NCERT-XII, Pg. # 112

174) NCERT Pg. # 201, 202

175) NCERT Pg. # 197, 198

176) NCERT Pg. # 211

177) Module # 84

178) NCERT Pg. # 206, 207

179) NCERT Pg. # 260

180) NCERT Page No. 219, 220